Daily Number Theory Drill #4

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Prime Factor Algebra · Difference of Squares · Bounding Factors.

Section A

1. Find the largest prime factor of 1001.

Ans: 13

2. What is the last digit of 2^{2026}?

Ans: 4

3. What is the remainder when $2^{10} + 3^{10}$ is divided by 5?

Ans: 3

4. Find the greatest common divisor of 120 and 84.

Ans: 12

5. How many positive integer factors does 144 have?

Ans: 15

6. What is the remainder when $11 \times 12 \times 13$ is divided by 10?

Ans: 6

7. What is the smallest positive integer divisible by $2, 3, 4, 5$, and 6?

Ans: 60

8. Find the sum of the prime factors of 210.

Ans: 17

9. What is the remainder when $10^{2026} - 1$ is divided by 9?

Ans: 0

10. Find the value of k if 2^k is the highest power of 2 dividing 20!.

Ans: 18

Section B

1. Find the smallest positive integer n such that $15n$ is a perfect square.

Ans: 15

2. Find all single digits d such that the four-digit number $5d72$ is divisible by 9.

Ans: 4

3. Find the number of trailing zeros in 50!.

Ans: 12

4. What is the remainder when 5^{2025} is divided by 7?

Ans: 6

5. If $x \equiv 3 \pmod 7$, find the remainder when $x^2 + 4x + 5$ is divided by 7.

Ans: 5

6. Find the smallest positive integer that leaves a remainder of 2 when divided by 3, and 3 when divided by 5.

Ans: 8

7. Find the sum of all positive integer divisors of 100.

Ans: 217

8. What is the last digit of $3^{101} + 7^{101}$?

Ans: 0

9. Find the number of positive divisors of $2^3 \times 3^4 \times 5$.

Ans: 40

10. Find the largest three-digit number that is a multiple of 11 and 13.

Ans: 858

Section C

1. Find all pairs of positive integers (x, y) such that $x^2 - y^2 = 45$.

Ans: $(23, 22), (9, 6), (7, 2)$

2. The prime number p can be expressed as $p = a^2 - b^2$, where a and b are also prime numbers. Find the value of p.

Ans: 5

3. Find the number of integer pairs (x, y) satisfying $x^2 - y^2 = 2018$. *(after TMUA 2018 Paper 1 Q14)*

Ans: 0

4. How many ordered pairs of positive integers (x, y) satisfy $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$?

Ans: 9

5. Find all integer values of x for which $\frac{x^2+5x+14}{x+3}$ is an integer.

Ans: $-11, -7, -5, -4, -2, -1, 1, 5$

6. How many divisors of 10^{10} are perfect squares?

Ans: 36

7. How many positive integers $n \leq 1000$ are not divisible by $2, 3$, or 5?

Ans: 266

8. Given that p and q are prime numbers satisfying $p = q^2 - 36$, find the value of p.

Ans: 13

Section D

1. Find all pairs of positive integers (m, n) such that $3^m - 2^n = 1$.

Ans: (1, 1), (2, 3)

2. For how many positive integers n is $n^4 + 4^n$ a prime number?

Ans: 1

3. A sequence is defined by $a_1 = 1$, $a_2 = 1$, and $a_{n+2} = a_{n+1} + a_n$. What is the remainder when a_{2026} is divided by 3?

Ans: 1

4. Let p, q, and r be prime numbers such that $p = 9q^2 - r^2$. Find the value of p.

Ans: 11

5. Find the largest positive integer n such that 2^n divides $3^{1024} - 1$.

Ans: 12